

CivicSync — Project Brief

Know Your Rights. Know What To Do Next.

Team: CivicSync **Theme:** Smart Automation

Problem Statement: Digital Assistant for Legal Awareness and Designing a KYR (Know-Your-Rights) Framework in India

1. Executive Summary

CivicSync is an evidence-grounded, multilingual legal-awareness assistant that helps Indian citizens understand a situation they're facing, find verified legal information relevant to it, and identify practical next steps — without needing to know legal terminology.

A citizen describes their problem in plain language. CivicSync interprets the situation, retrieves relevant verified evidence, checks how well that evidence actually supports an answer, and only then generates a structured, source-backed response.

CivicSync is not an AI lawyer and does not provide legal advice. It is a legal-awareness and citizen-guidance system: it helps people understand their rights and options, and points them toward the right authorities, documents, and next steps.

The web platform (React/TypeScript frontend, FastAPI backend, RAG pipeline over a pgvector knowledge base) is built and tested — 254 automated tests passing, 0 failing. A voice-helpline extension for low-connectivity citizens is a proposed future direction, not a built feature.

2. Problem

- Most citizens do not know their basic legal rights, or which authority/process applies to their situation.
- Legal language is dense and inaccessible to non-experts.
- Existing chatbots tend to generate generic or unverified answers, which is especially risky in a legal context.
- Language is a barrier: most legal-awareness tools exist only in English.
- Rural and low-connectivity citizens are largely excluded from digital legal-awareness tools.

3. Proposed Solution

CivicSync converts a citizen's plain-language query into a structured, actionable, source-backed response through four stages:

- Understand the situation (domain, issue, intent).
- Retrieve relevant evidence from a verified legal knowledge base.
- Qualify that evidence — decide how much it actually supports an answer.
- Generate a citizen-friendly response, strictly constrained by the qualified evidence.

The guiding principle: evidence controls what the AI is allowed to explain. The LLM is used to explain retrieved, qualified evidence in plain language — not to generate legal conclusions from memory.

4. How CivicSync Works (End-to-End Flow)

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Citizen Query (English / Hindi / Gujarati)
  → Situation Understanding (Domain / Issue / Intent / Confidence)
  → Vector Embedding (multilingual-e5-base, 768-dim)
  → pgvector Similarity Retrieval → Relevant evidence chunks
  → Evidence Qualification (Supported / Partially Supported / Insufficient)
  → Action Engine (structures qualified context)
  → Constrained LLM Generation → Structured citizen response
  → Sources + Disclaimer displayed to citizen
```

Example — real interaction

Citizen: "My employer has not paid my salary for two months. What should I do?"

- Situation identified as: labour / non-payment of wages
- Relevant evidence retrieved from the knowledge base
- Evidence qualified (supported / partially supported / insufficient)
- Any missing information flagged (e.g., is this formal or informal employment?)
- Practical next steps generated
- Relevant authority/channel and required documents identified
- Sources and disclaimer displayed alongside the answer

Note: CivicSync does not invent specific deadlines, section numbers, or legal provisions — it only surfaces what is present and qualified in its retrieved evidence.

5. Technical Architecture

Frontend

- React + TypeScript + Vite
- Responsive, accessible conversational UI
- English / Hindi / Gujarati localization (i18next / react-i18next)
- Browser-based Speech-to-Text and Text-to-Speech (where supported)
- Action-first response layout with citizen-friendly loading, error and retry states

Backend

- FastAPI service exposing POST /api/query
- Modular CivicSync reasoning pipeline (situation understanding → retrieval → qualification → action → generation)
- Provider abstraction layer for LLMs, decoupling the pipeline from any single vendor

Vector Database

- Supabase (Postgres) with the pgvector extension
- Embedding model: multilingual-e5-base (768 dimensions)

- Knowledge base: ~993 chunks currently indexed (includes an Insurance V2 set of ~647 regulatory chunks)

LLM Layer

- Provider abstraction currently supports Groq, Gemini, and Ollama
- The LLM's role is narrow and disciplined: explain qualified evidence in plain language, not generate answers from unconstrained internal knowledge

6. Why RAG, pgvector, and Evidence Qualification

Why RAG (Retrieval-Augmented Generation)?

- Legal information must be grounded in real source material, not model memory.
- Laws and regulations are domain-specific and change over time — an LLM's static training data cannot be trusted alone.
- Retrieving evidence first means every part of the answer can be traced back to a source document.

What pgvector does, simply put

It stores numerical representations ("embeddings") of legal text chunks and finds the chunks that are semantically closest in meaning to the citizen's query — even if the citizen didn't use the exact legal wording.

Why Evidence Qualification matters

Retrieved evidence is never automatically trusted. Each piece of evidence is classified as:

- SUPPORTED — evidence clearly answers the query
- PARTIALLY_SUPPORTED — evidence is related but incomplete
- INSUFFICIENT — evidence does not adequately cover the query

If evidence is insufficient, CivicSync asks a clarifying question or plainly states the limitation — it does not fabricate an answer. This is what we mean by evidence-gated generation: it reduces unsupported claims, though it is not an absolute guarantee against every possible error.

7. Multilingual and Voice Accessibility

- UI available in English, Hindi, and Gujarati; language selection updates UI text and is passed through the existing API contract.
- Voice locale mapping: English → en-IN, Hindi → hi-IN, Gujarati → gu-IN.
- Voice input is transcribed into editable text before submission, so the citizen always stays in control of what is actually sent.

Accessibility (built)

- Semantic HTML and keyboard navigation
- Visible focus states
- Screen-reader-friendly labels and aria-live announcements

- Accessible loading/error states
- Responsive design across mobile, tablet, and desktop

8. Rural / Low-Connectivity Extension — Proposed Roadmap, Not Built

Many rural citizens have basic cellular calling even where reliable mobile internet is unavailable. We are exploring a voice-helpline model to extend CivicSync's reach.

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Citizen (ordinary phone, calling)
  → CivicSync Voice Helpline
  → Speech Recognition / IVR Layer
  → Existing CivicSync Reasoning Pipeline
  → RAG + Evidence Qualification
  → Simple Voice Response
  → Human / Legal Escalation (when necessary)
```

Important clarification: the citizen's phone would not need mobile internet in this model — but CivicSync's backend and telephony infrastructure would still require connectivity on the service side.

Proposed safeguards

- Conversation/session recovery after call drops
- Speech confidence checks and clarification prompts when speech is unclear
- Emergency-case escalation
- Minimal data retention
- Human escalation for complex or high-risk situations

9. Security, Privacy and Safety

- Voice input is converted to editable text before submission — no silent auto-submission.
- Minimal data retention is a stated design principle for any future telephony layer.
- Evidence qualification prevents the system from presenting unsupported answers as certain.
- Clear disclaimers accompany every response; the system is explicitly not a substitute for a lawyer, authority, or emergency service.
- High-risk or ambiguous cases are designed to route toward human/legal escalation rather than an automated final answer.

10. Cost Optimization

CivicSync is intentionally kept architecturally simple. Planned/possible cost controls (future direction):

- Response caching for frequently asked questions
- Simple model routing — smaller/cheaper models for simple tasks, stronger models only when needed
- Provider abstraction, so pricing/performance can be balanced across Groq, Gemini, and Ollama
- Retrieval-before-generation, which reduces unnecessary LLM calls
- Basic request/latency/token monitoring

We deliberately avoid unnecessary complexity such as multi-agent swarms, fine-tuning pipelines, Kubernetes orchestration, or multiple agent frameworks — these are not part of the current design and are not planned near-term.

11. Innovation / Differentiators

- Evidence-grounded, not a generic chatbot — every response traces back to retrieved, qualified evidence.
- Situation understanding before guidance — the system classifies domain/issue/intent before retrieving anything.
- Evidence qualification before generation — the LLM only explains evidence that has passed a supported/partially-supported/insufficient check.
- Action-first citizen guidance — outputs are structured into rights, immediate actions, authorities, and required documents, not a wall of text.
- Trilingual accessibility — English, Hindi, and Gujarati, with voice support in all three.
- Designed for future low-connectivity access — web today, voice-helpline roadmap for citizens without reliable internet.
- Traceable sources and honest limitations — sources and disclaimers are shown with every answer.

12. Impact

- Makes legal information easier to understand for non-experts
- Reduces the jargon barrier between citizens and legal knowledge
- Helps citizens identify practical, concrete next steps
- Improves access for Hindi and Gujarati speakers
- Enables voice-based interaction for citizens less comfortable typing
- Provides source-traceable legal awareness rather than opaque answers
- Lays a foundation for a broader Know-Your-Rights infrastructure in India

13. Roadmap — Next & Future

NEXT	FUTURE
• Telephony/voice helpline prototype • Better rural/low-connectivity access • More Central and State legal sources • More Indian languages • Improved speech handling • Cost-aware model routing and caching • Human/legal referral integration	• Wider national legal-awareness infrastructure • Public voice/telephony access • Broader language and regional coverage • Continuous source/version updating

CivicSync does not replace lawyers, government authorities, or emergency services — it is a guidance and awareness layer, not a decision-maker.